Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building **optimization-focused backend systems**, **Python data pipelines**, and user-facing products, with C++/systems fundamentals, algorithmic problem-solving, and interest in climate-impact software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized multi-tenant backend workflows supporting 500+ users, reducing latency from **1.2s to 350ms** through algorithmic query tuning, indexed access patterns, and structured service improvements.
- Built ETL-style integration pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, improving reliability and accelerating diagnosis of data and runtime anomalies.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed full-stack React and Flask features handling 10,000+ daily requests, translating user feedback into responsive interfaces and maintainable APIs while improving usability, testing discipline, and release confidence.
- Designed concurrent batch-processing workflows using multithreading and secure file ingestion, processing 1K+ files per batch while reducing feedback turnaround by **60%** under production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python experimentation pipelines across 100,000+ trajectory records, enabling reproducible evaluation, simulation analysis, and benchmark consistency for model-driven planning and optimization workloads.
- Produced technical reports and targeted validation scenarios translating findings into actionable improvements, strengthening design reviews, reducing handoff ambiguity, and supporting better engineering decisions across collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable data movement and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence during collaborative release cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based operations platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services and responsive user-facing interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

Skills

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, Bash; Rust familiarity
Systems & Algorithms: optimization problems, data structures, operating systems, concurrency, simulation workflows, performance debugging
Backend & Data: FastAPI, Flask, REST APIs, ETL pipelines, PostgreSQL, SQL Server, MongoDB, data validation
Cloud & Delivery: AWS, Azure, Docker, Kubernetes, Git/GitHub, Jenkins, CI/CD, user-focused product iteration

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling with **C++** logic, React controls, and Gantt visualization, improving algorithm debugging efficiency by **20%** across simulated workloads.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a containerized RESTful data processing API with automated CI/CD releases and validation checks, improving deployment reliability, interoperability, and supportability for production data workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, maintaining Java and TypeScript modules that accelerated schema exploration, query authoring, and structured debugging of data workflows.